Highlights

Field accuracy is not heat accuracy: wall-flux-constrained surrogates for non-Boussinesq convection

- On 200 held-out classical-cavity samples, 0.09% field error coexists with 2.0% integral Nu error.
- A 22-sample random-wall replication gives 0.47% field error and 4.04% Nu error.
- A wall-flux penalty cuts both-wall Nu error to 0.01% (classical) and 0.58% (random walls).
- Boussinesq–compressible gap factorises as $D = \varepsilon P(X)$ over 627 pairs.
- External ε factorisation is reusable; rescaled plain matches within 0.2%.